

Mathis Petrovich

Research Scientist



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Summary I am an Research Scientist at NVIDIA in the Spatial Intelligence Lab working on 3D human motion synthesis. I was previously an ELLIS PhD student in the IMAGINE computer vision team of École des Ponts ParisTech (ENPC) and in the Perceiving Systems Department of Max Planck Institute for Intelligent Systems (MPI-IS) advised by Gül Varol (ENPC) and Michael J. Black (MPI).

Google Scholar <https://scholar.google.com/citations?user=WUKF2ZwAAAA>

Research Experience

2024 - : Research Scientist, NVIDIA, Switzerland

Subject: Motion Foundation Model and applications

Team: Spatial Intelligence lab

2020 – 2024: PhD student, ENPC/MPI, France/Germany

3.5 years

Subject: Natural Language Control for 3D Human Motion Synthesis

Advisors: Gül Varol and Michael J. Black

2023: Research Intern, NVIDIA, Switzerland

5.5 months

Subject: Multi-Track Timeline Control for Text-Driven 3D Human Motion Generation

Advisors: Xue Bin Peng and Davis Rempe

2019 - 2020: Research Intern, RIKEN AIP, University of Kyoto, Japan

9 months

Subject: Machine learning and optimal transport

Advisor: Makoto Yamada

2019: Research Intern, DxO Labs, France

6 months

Subject: Semantic segmentation and image matting

Advisor: Wolf Hauser

2018: Research Intern, Carnegie Mellon University, United States

5 months

Subject: Object tracking in videos

Advisor: Martial Hebert

2017: Research Intern, LIF, France

2 months

Subject: Correction strategy for natural language parser

Advisor: Alexis Nasr

Education

2020 - 2024: ENPC/MPI, PhD student

France/Germany

Natural Language Control for 3D Human Motion Synthesis

2016 - 2020: École Normale Supérieure (ENS) Paris-Saclay, MSc

Paris, France

Research engineering school, theoretical and applied computer science

- 2018 - 2019: ENS Paris-Saclay, Master 2 / Master MVA: machine learning and computer vision

- 2017 - 2018: ENS Paris-Saclay, Master 1 / Master of research in theoretical computer science (MPRI)

- 2016 - 2017: Diderot University, BSc / Theoretical computer science

2014 - 2016: Lycée Masséna, MPSI/MP*

Nice, France

Preparation course for exams to enter French engineering schools

Academic Activities

Recent Publications

- **2024, CVPRW: Mathis Petrovich**, Or Litany, Umar Iqbal, Michael J. Black, Gül Varol, Xue Bin Peng, Davis Rempe
STMC: Multi-Track Timeline Control for Text-Driven 3D Human Motion Generation.
- **2023, ICCV: Mathis Petrovich**, Michael J. Black, Gül Varol
TMR: Text-to-Motion Retrieval Using Contrastive 3D Human Motion Synthesis.
- **2023, ICCV: Nikos Athanasiou*, Mathis Petrovich***, Michael J. Black, Gül Varol
SINC: Spatial Composition of 3D Human Motions for Simultaneous Action Generation.
- **2022, 3DV: Nikos Athanasiou, Mathis Petrovich**, Michael J. Black, Gül Varol
TEACH: Temporal Action Composition for 3D Human.
- **2022, ECCV (Oral): Mathis Petrovich**, Michael J. Black, Gül Varol
TEMOS: Generating diverse human motions from textual descriptions.
- **2021, ICCV: Mathis Petrovich**, Michael J. Black, Gül Varol
ACTOR: Action-Conditioned 3D Human Motion Synthesis with Transformer VAE.
- **2022, ECML: Mathis Petrovich***, Chao Liang*, Ryoma Sato, Yanbin Liu, Yao-Hung Hubert Tsai, Linchao Zhu, Yi Yang, Ruslan Salakhutdinov, Makoto Yamada
FROT: Feature Robust Optimal Transport for High-dimensional Data.

Reviewing

- Computer Vision and Pattern Recognition (CVPR) 2023 (**Outstanding Reviewer**), 2024
- International Conference on Computer Vision (ICCV) 2023
- European Conference on Computer Vision (ECCV) 2022, 2024
- Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2022, 2023
- International Journal of Computer Vision (IJCV) 2022, 2023
- International Conference on 3D Vision (3DV) 2024
- SIGGRAPH 2023
- Eurographics 2024
- Computers & Graphics 2021

Teaching

- 2021 - 2024: ENS Paris-Saclay, *Teaching Assistant*, M2 (Master MVA)
Object recognition and computer vision (RecVis)
- 2022 - 2023: ENPC Engineering school, *Project supervisor*
Research project on computer vision
- 2020 - 2021: ENPC, *Teacher*, L3 (Bachelor)
Introduction to programming, in C++ (1PROG)

Recent open-source repositories

 nv-tlabs/stmc	130 ★	8 📄
 Mathux/TMR	261 ★	22 📄
 Mathux/TEMOS	426 ★	24 📄
 Mathux/ACTOR	414 ★	56 📄

Miscellaneous

Research interests

- 3D human motion modeling
- Generative models
- Robotics
- Computer vision
- Machine learning

Languages

-  **French:** Native speaker
-  **English:** C1 Level (IELTS Band 7)
-  **German:** A2

References

- **Gül Varol:** gul.varol@enpc.fr
- **Michael J. Black:** black@tuebingen.mpg.de